HARMONIC BALANCE FINITE ELEMENT ANALYSIS OF THIRD ORDER INTERMODULATION PRODUCTS IN FERRITE DEVICES

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Abstract

Results of a study of nonlinearities and intermodulation products for 3-port H-plane waveguide circulators are presented, based on a harmonic balance finite element analysis of the third-order nonlinearity in ferrites. Numerical results are derived for the field strength and the power level of the intermodulation signal.

Index Terms - Finite elements, harmonic analysis, ferrite, intermodulation, junction circulator, nonlinear interaction.

I. INTRODUCTION

Waveguide devices are widely used in high-power applications thanks to their low-losses. In high-power applications for waveguide devices with ferrite inserts, the motion of the magnetization vector can no longer be described by a linear set of equations. Considering higher-order terms of the permeability, echo signals in the time domain and intermodulation coupling in the frequency domain must be included in the nonlinear theory. In particular t5hird-order nonlinearity in ferrites generate intermodulation noise in the signal bandwidth.

In a frequency domain analysis, conventional single-harmonic (SH) finite elements, does not allow for the computation of signal coupling between an in-band signal and a strong out-of-band interferer. A multiple-harmonics (MH) analysis would be an attractive alternative to time domain analysis.

The theory of harmonic balance finite element (HBFE) method, also known as multiharmonic finite element, is presented for the analysis of a generic *H*-plane microwave ferrite device. This technique has been successfully employed in magnetostatics problems [2]-[3] but not to high frequency ones. The final goal is to analyze a ferrite device affected by nonlinear behavior.

II. FORMULATION

Consider a waveguide circulator (Fig. 1). Due to the *H*-plane symmetry of the actual 3D device, a complete solution is attained by solving the associated 2D problem just for the out-of-plane component of the electric field (E_z)

$$\nabla \times [\mu_r]^{-1} \nabla \times (E_z \hat{\mathbf{z}}) + k^2 \varepsilon_r (E_z \hat{\mathbf{z}}) = 0 \qquad (1)$$

within the domain Ω, subdomain Ω_1 is fully linear while subdomain Ω_2 contains the nonlinear ferrite. Perfect electric conductor boundary conditions holds on Γ_0 while modal expansions are enforced at ports Γ_i , i=1…3. k is the free space wavenumber, ε_r is the isotropic permittivity and $[\mu]$ has the form

$$[\mu]=\begin{bmatrix} \mu & -jk_r & 0 \\ jk_r & \mu & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{cases} \mu = 1+\dfrac{\omega_0\omega_m}{\omega_0^2-\omega^2} \\ k_r = -\dfrac{\omega\omega_m}{\omega_0^2-\omega^2} \end{cases}, \quad \omega_0 = \gamma H_0,\ \omega_m = \gamma M_s, \tag{2}$$

where ω is the angular frequency, γ the gyromagnetic ratio, H_0 the internal DC magnetic field and M_s the saturation magnetization. The conventional finite element formulation for anisotropic media can be retrieved in [4].

The diagonal permeability tensor coefficients can be expressed as

$$\left[\mu\left(E_z(\mathbf{r}),\mathbf{r}\right), k_r\left(E_z(\mathbf{r}),\mathbf{r}\right)\right]=\begin{cases} [1,0] & \mathbf{r}\in\Omega_1 \\ [\bar{\mu},\bar{k}_r]+\chi\left(E_z(\mathbf{r})\right) & \mathbf{r}\in\Omega_2 \end{cases} \tag{3}$$

where $E_z(\mathbf{r})$ is the electric field in the generic point $\mathbf{r}$ and $\chi\left(E_z(\mathbf{r})\right)$ is a quadratic scalar operator describing the nonlinear behavior.

Considering a two-tones problem, the MH behavior is introduced by approximating the electric field as

$$\begin{aligned}\hat{E}_z=&e_1\cos(\omega_1 t)+e_2\cos(\omega_2 t)+e_3\cos((2\omega_1-\omega_2)t)+e_4\cos((2\omega_2-\omega_1)t)+\\ &+e_5\cos((2\omega_1+\omega_2)t)+e_6\cos((2\omega_2+\omega_1)t)+e_7\cos(3\omega_1 t)+e_8\cos(3\omega_2 t)+\ldots\end{aligned} \tag{4}$$

where ω_1 is the in-band signal angular frequency and ω_2 is the angular frequency of the edge-band interferer, and the hat, here and in the following, denotes approximation deriving from the truncation of the series.

The complex harmonic amplitudes constitute the unknowns of a conventional finite element system, with k in (1) computed with the corresponding harmonic frequency

$$k_1=\frac{\omega_1}{c_0},\quad k_2=\frac{\omega_2}{c_0},\quad k_3=\frac{2\omega_2-\omega_1}{c_0},\quad k_4=\frac{2\omega_1-\omega_2}{c_0},\ldots \tag{5}$$

being c_0 the speed of light in free space.

The harmonics retained in (4) are selected according to the quadratic operator $\chi(.)$, which generates third order intermodulation products, the most critical being ($2\omega_1$ - ω_2) and ($2\omega_2$ - ω_1). Eight harmonic amplitudes in (4) will be considered.

The nonlinear behavior of the relative permeability tensor coefficients generates coupling between the previous harmonic signals at the frequencies that expand the electric field. The dependence can be explicitated on the magnetic field as:

$$\hat{\mu}_r = \bar{\mu}_r + \chi(E_z) = \bar{\mu}_r + \alpha\left(\left|\hat{H}_x(E_z)\right|^2 + \left|\hat{H}_y(E_z)\right|^2\right) \qquad (6)$$

which constitutes a wider sum of harmonics than that of the field, α being a constant. Quadratic permeability tensor coefficients leads to the following expression

$$\hat{\mu}_r = \mu_{r_0} + \sum_{m+n>0 \text{ even}} \mu_{r^+_{m,n}} \cos(m\omega_1 t + n\omega_2 t) + \mu_{r^-_{m,n}} \cos(m\omega_1 t - n\omega_2 t) \qquad (7)$$

The harmonic amplitude μ_r of the permittivity is computed by testing its time evolving value (7) at the sought frequency

$$\mu_{r_0} = \frac{\omega}{2\pi}\int_0^{\frac{2\pi}{\omega}} \bar{\mu}_r + \alpha\left(\left|\hat{H}_x\right|^2 + \left|\hat{H}_y\right|^2\right) dt,$$

$$\mu_{r^\pm_{m,n}} = \frac{GCD(m\omega_1, n\omega_2)}{\pi}\int_0^{\frac{2\pi}{GCD(m\omega_1,n\omega_2)}} \left(\bar{\mu}_r + \alpha\left(\left|\hat{H}_x\right|^2 + \left|\hat{H}_y\right|^2\right)\right)\cos(m\omega_1 t \pm n\omega_2 t)\, dt, \qquad (8)$$

being GCD the greater common divisor function.

Harmonic testing is then performed, leading to a large scale linear HBFE system which has formally the same structure as a conventional FE system, the nonlinearity being tackled via a Picard iteration.

III. NUMERICAL RESULTS

The device analyzed is sketched in fig. 1, the ferrite parameters are those presented in [5]. For a linear ferrite the devices operates correctly as shown in Fig. 2: Devices is matched at 10GHz and isolation (S31) is better than -15dB. In the case of a nonlinear ferrite, with $\alpha = 2\times 10^{-7}\,\mathrm{m^2A^{-2}}$, analysis is performed with an impinging signal at $\omega_1 \in [2\pi 8\times 10^9, 2\pi 12\times 10^9]$ with an impinging power of (a) 0.5W; (b) 0.75W; (c) 1W; and an interferent signal at $\omega_2 = 2\pi 7.8\times 10^9$ with a corresponding power level of (a) 50W; (b) 75W; (c) 100W. Graph clearly shows the effect of the nonlinearity both in a frequency shift and in the presence of an output power level at the third intermodulation product.

IV. CONCLUSION

A HBFE approach for waveguide devices containing nonlinear ferrites has been presented. The proposed method allows for the prediction, in frequency domain, of the amplitude of the intermodulation products and relative signal degradation.

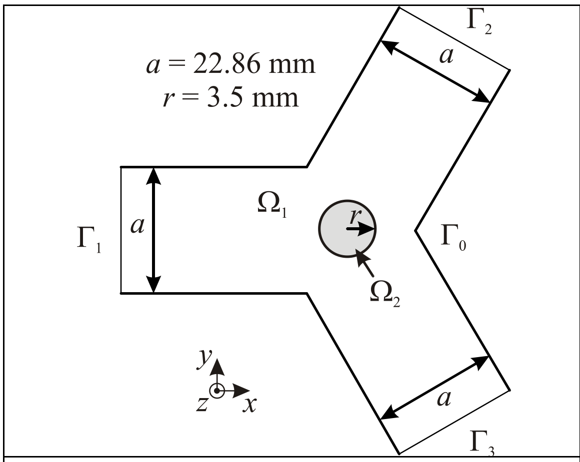

FIG. 1 - Sketch of the analyzed circulator H-plane cross-section. A cylindrical ferrite post of radius r is placed in the middle of the device.

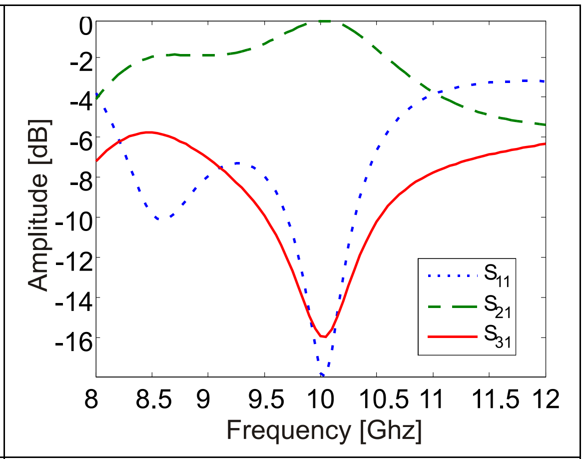

FIG. 2 - Amplitude of the three main scattering parameters for the linear case, for an impinging TE_{10}.

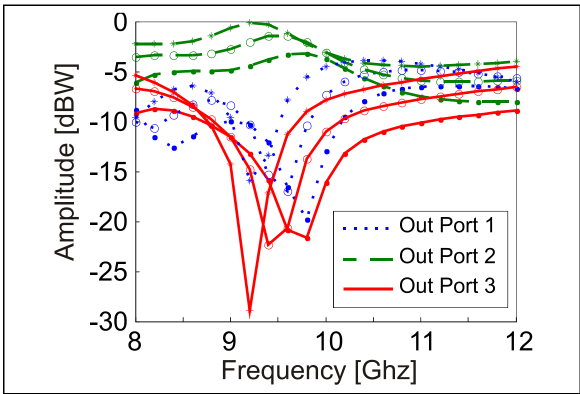

FIG. 3 - Amplitude of the output power at w_1 nonlinear case. (a) bullets (b) empty circles (c) asterisks.

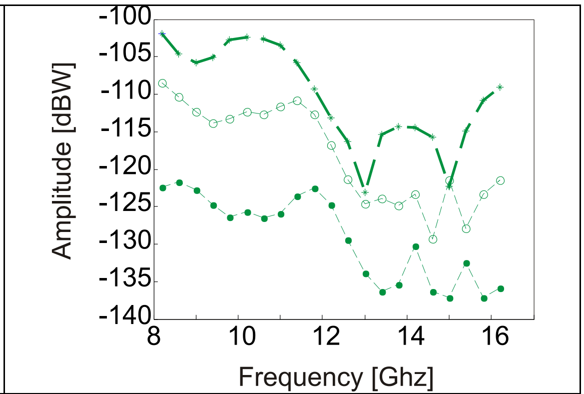

FIG. 4 - Amplitude of the output power at port 2 for the third harmonic $2w_1 - w_2$ for the nonlinear case. (a) bullets (b) empty circles (c) asterisks.

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